

Aldehyde Dehydrogenase-1 Expression Predicts Unfavorable Outcomes in Patients with Esophageal Squamous Cell Carcinoma

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Abstract. *Background: Aldehyde dehydrogenase-1 (ALDH1) has been shown to be a potential cancer stem cell marker in different types of cancer. However, the role of its expression in tumor cells and the microenvironment in different types of cancer is still controversial. Materials and Methods: ALDH1 protein was immunohistochemically investigated and analyzed in 157 samples of surgically dissected esophageal squamous cell carcinoma (ESCC) tissues. Results: ALDH1 protein expression in ESCC tumor cells was significantly associated with poor differentiation ($p < 0.05$) and strongly positive ALDH1 expression in tumor cells was related with shorter overall survival ($p < 0.05$), while the expression of ALDH1 in ESCC stromal cells had no significant relationship with clinicopathological features ($p > 0.05$). Conclusion: High expression of cancer stem cell marker ALDH1 in ESCC cells may thus portend a poor prognosis. However, its expression in the tumor microenvironment did not appear to have a role in ESCC behavior. More studies are warranted to find out the mechanisms to explain this.*

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Esophageal squamous cell carcinoma (ESCC) is a highly malignant lesion and is the fourth-leading cause of cancer-associated death in China (1). Despite improved surgical standards and the application of adjuvant treatment for resectable tumors, survival rates in esophageal cancer remain low (2). More powerful prognostic biomarkers are needed to define the risk profiles of patients with ESCC and new therapies that target chemo- and radioresistance are needed in order to improve the survival rate.

Many studies have repeatedly confirmed the existence of cancer stem cells (CSCs) (3-5). CSCs represent only a small proportion of all tumor cells and are not treatable by current chemotherapy or radiotherapy. They have been implicated in tumorigenesis, relapse, and metastasis of cancer (6, 7). Hence, identifying CSC markers in ESCC samples may provide important information regarding patient prognosis and response to therapy (8, 9).

The aldehyde dehydrogenase (ALDH) family, a large superfamily of proteins, catalyzes dehydrogenation of aldehydes to their corresponding carboxylic acids, and can be found in the cytosol, nucleus, and endoplasmic reticulum (10). One of these family members, ALDH1, consists of six subtypes, three of which (ALDH1A1, ALDH1A2 and ALDH1A3) catalyze the oxidation of retinaldehyde to retinoic acid, an important regulator of gene expression (11). ALDH1 takes part in regulating cell differentiation, proliferation and motility. Its regulatory role is through the retinoid signaling pathway. Li *et al.* reported that inhibiting ALDH1-mediated retinoid signaling impairs human fetal

Table I. The criteria of the Allred scoring system (47) used for evaluating aldehyde dehydrogenase-1 expression in esophageal squamous cell carcinoma cells and stromal cells in our study.

1. The criteria of the intensity scoring system				
Intensity score	0	1	2	3
Staining intensity	Negative	Weak	Moderate	Strikingly positive at low magnitude
2. The criteria of percentage scoring system				
Percentage score	0	1	2	3
Stained cells	0	<25%	25%-50%	>50%
3.				
Total score ^a	0	1-4	5-6	
	(-)	(+)	(++)	

^aThe total score was obtained by adding the percentage score to the intensity score. It ranges from 0 to 6.

islet cell differentiation and survival (12), indicating that ALDH1 may be involved in cancer genesis and development.

Studies have shown that ALDH1 is a marker for CSCs in a variety of adult cancer types, including breast, gastrointestinal, ovarian and lung among others (13-18). CSC markers have been associated with increased aggressiveness in some tumor types. The predictive role of ALDH1 in cancer is unclear, with some studies showing ALDH1 expression being associated with poor overall survival and others showing contrasting results (14, 15, 18-20).

In the present study, we immunohistochemically examined the expression of ALDH1 in the tumor and stromal cells of human ESCC tissue and explored its relationship with conventional clinicopathological characteristics in a series of 157 patients with ESCC (Anyang, China) with long-term follow-up.

Materials and Methods

Patients. In total, 157 patients, including 95 men and 62 women, were included in this retrospective study. All patients underwent surgery with curative intent after a diagnosis of ESCC between 1989 and 1994 at the Anyang Tumor Hospital, Henan, China. All patients were followed-up from the confirmed diagnosis date until death or 31 May 2004, except for 19 patients, who were lost to follow-up (Table I). Tumors were classified according to the 2003 standards established by the Union for International Cancer Control (21). The 157 surgically removed samples were routinely fixed in formalin, processed, and embedded in paraffin blocks for diagnosis and research use. This project was completed at Anyang Tumor Hospital and the Norwegian Radium Hospital. The Ethics Committee of Anyang Tumor Hospital and Anyang Hygiene Bureau approved this study. All involved patients gave written consent for participation in this research, and all consent forms were filed at the Anyang Tumor Hospital, Henan, China.

Immunohistochemistry. The Dako EnVision™ Flex+ (K8012; Dako, Glostrup, Denmark) was applied for immunohistochemical staining. De-paraffinization and unmasking of epitopes were performed in PT link with low pH target retrieval solution (Dako), followed by

blocking peroxidase for 5 min with DAKO blocking buffer. The slides were then incubated with the following reagents: mouse antibody to human ALDH1 (1:3000, 83 ng IgG1/ml, Clone 44; BD Transduction Laboratories™ Oslo, Norway) at 4°C overnight, EnVision™ Flex+ mouse linker for 15 min and EnVision™ Flex+HRP for 30 min at room temperature. All sections were stained with 3, 3'-diaminobenzidine tetrahydrochloride for 5 min and counterstained with hematoxylin. Finally, all slides were dehydrated and mounted in Richard-Allan Scientific Cyto seal XYL (Thermo Scientific, Waltham, MA, USA). Human liver tissue is always positive for ALDH1 staining, was therefore used as a positive control in this study; negative controls were performed using the same concentration of non-immune mouse IgG instead of mouse anti-human ALDH1.

Immunohistochemical scoring system. Immunostaining was evaluated by two pathologists from the Norwegian Radium Hospital, Oslo University Hospital, Oslo, Norway, and they were blinded to clinical information. A third observer re-evaluated and confirmed the slides that had discordant diagnoses. Semiquantitative evaluation of immunohistochemical staining was used for evaluating ALDH1 expression levels in the ESCC samples. Specimens were reviewed for staining intensity and staining extent. The intensity of the immunohistochemical staining was measured on a scale of 0 to 3. Staining extent was rated according to the percentage of positively stained cells in the field. Samples with no staining of cells were rated as 0, those with <25% of cells stained as 1, those with 25 to 50% of cells stained as 2, >50% of cells stained were rated as 3. The sum of the intensity score and the extent score was expressed as a total score (Table I). The slides were grouped as negative or weakly positive (-), moderately positive (+) and strongly positive (++) when the total score was 0, 1 to 4, and 5 to 6, respectively.

Statistical analysis. Statistical analysis was performed using SPSS statistical software package, Version 17.0 (SPSS, Chicago, IL USA). Chi-square tests (Pearson and linear-by-linear as appropriate) were performed to analyze the associations between ALDH1 expression and clinicopathological features. Survival curve was plotted through the Kaplan-Meier method and compared with the use of the two-sided log-rank test. The Cox proportional regression hazard model was used in multivariate analysis to identify the significant factors that were correlated with prognosis. For all analyses, a *p*-value less than 0.05 was regarded as statistically significant.

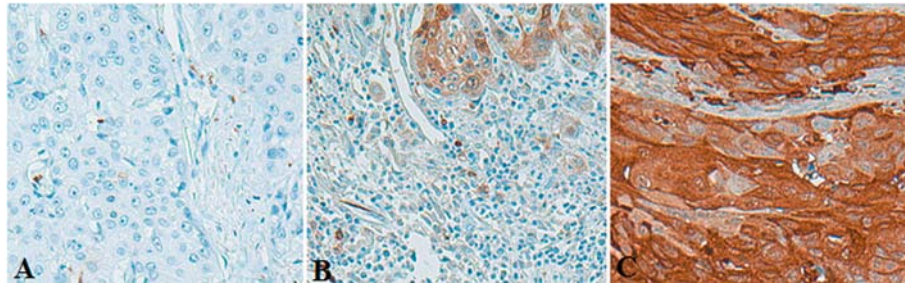


Figure 1. Immunohistochemical staining of aldehyde dehydrogenase-1 (ALDH1) in esophageal squamous cell carcinoma. The typical diffuse cytoplasmic staining of the protein can be seen in esophageal squamous cell carcinoma. A: ALDH1 negativity; B: weak ALDH1 positivity; C: strong ALDH1 positivity (magnification $\times 200$).

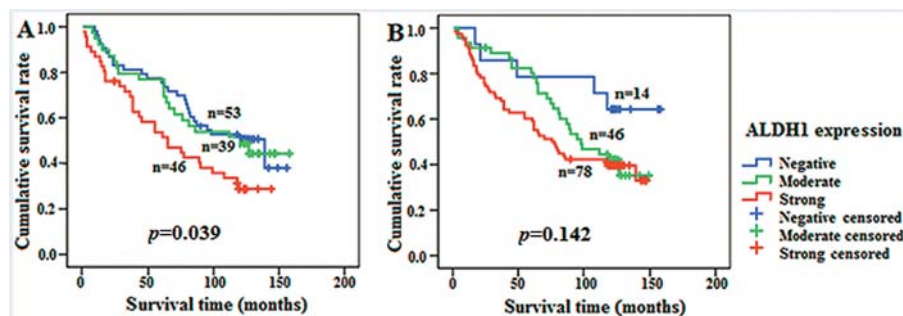


Figure 2. Correlation between aldehyde dehydrogenase-1 (ALDH1) expression and prognosis of patients with esophageal squamous cell carcinoma. A: Survival plot shows that high ALDH1 expression in tumor cells in esophageal squamous cell carcinoma is associated with poorer overall survival than low and negative expression ($p < 0.05$, Kaplan–Meier). B: The expression level of ALDH1 in stromal cells in esophageal squamous cell carcinoma has no significant association with overall survival probability ($p > 0.05$, Kaplan–Meier).

Results

ALDH1 was variably detected in esophageal squamous cell carcinoma specimens. Immunoreactivity to ALDH1 was variably detected in both tumor and stromal cells in all the ESCC samples (Figure 1). Immunostaining was limited to the cytoplasm and cell membrane. Out of the total 157 samples, 62 cases (39.49%) were negative, 42 (26.75%) were moderately positive, and 53 (33.76%) were strongly positive for ALDH1 in tumor cells (Table II). Compared to tumor cells, ALDH1 expression in the stromal cells was generally rather strong. Negative, moderately positive, and strongly positive expression of ALDH1 in the stromal cells was observed in 16 (10.19%), 53 (33.76%), and 88 (56.05%) cases, respectively (Table III).

Clinicopathological variables in 157 cases of esophageal squamous cell carcinoma. The association between ALDH1 protein expression in tumor cells and stromal cells and the clinicopathological features is summarized in Table II. ALDH1 expression in ESCC tumor cells was significantly

positively associated with differentiation grade ($p < 0.05$). Strong ALDH1 staining was noted in 25/53 (47%) poorly differentiated ESCC samples, 22/48 (46%) moderately differentiated samples, but only in 6/62 (10%) well-differentiated samples. However, no significant association was found between ALDH1 protein expression and clinical parameters such as age, sex, tumor size, and TNM stage. As summarized in Table III, there was no significant difference between ALDH1 expression in stromal cells according to clinicopathological features of ESCC.

Survival analysis. As 19 patients were lost to follow-up, our survival analysis included a total of 138 cases. The median overall survival was 90 months with a 95% confidence interval (CI) of 61.197 months to 118.803 months. For the 138 patients, ALDH1 expression in tumor cells was negative in 53 cases, moderate in 39 cases, and strong in 46 cases. Kaplan–Meier survival curves and the log-rank test demonstrated that tumor ALDH1 expression in ESCC was significantly associated with poor overall survival ($p = 0.039$; Figure 2A). The median overall survival for the tumor

Table II. Clinicopathological information of patients with esophageal squamous cell carcinoma and their relationship with expression of aldehyde dehydrogenase-1 (ALDH1) in tumor cells.

Variable	Total	ALDH1 expression			p-Value
		(-)	(+)	(++)	
Age (years)					0.640
<50	41	15	11	15	
50-60	57	24	13	20	
>60	59	23	18	18	
Gender					0.983
Male	95	38	25	32	
Female	62	24	17	21	
Tumor size (mm)					0.383
<30	35	15	7	13	
30-60	107	38	32	37	
>60	15	9	3	3	
TNM stage					0.293
I/II	105	44	29	32	
III/IV	52	18	13	21	
Differentiation grade					0.000
Well	58	31	21	6	
Moderate	48	15	11	22	
Poor	51	16	10	25	

Negative or mildly positive (-), moderately positive (+) and strongly positive (++)

Table III. Clinicopathological information of patients with esophageal squamous cell carcinoma and their relationship with expression of aldehyde dehydrogenase-1 (ALDH1) in stromal cells.

Variable	Total	ALDH1 expression			p-Value
		(-)	(+)	(++)	
Age (years)					0.524
<50	41	5	9	27	
50-60	57	5	23	29	
>60	59	6	21	32	
Gender					0.406
Male	95	9	30	56	
Female	62	7	23	32	
Tumor size (mm)					0.640
<30	35	1	16	18	
30-60	107	15	32	60	
>60	15	0	5	10	
TNM stage					0.392
I/II	105	9	36	60	
III/IV	52	7	17	28	
Differentiation grade					0.813
Well	58	7	16	35	
Moderate	48	6	17	25	
Poor	51	3	19	29	

Negative or mildly positive (-), moderately positive (+) and strongly positive (++)

Table IV. Multivariate analysis of overall survival in a total of 138 patients with esophageal squamous cell carcinoma.

Variable	RR	95% CI	p-Value
ALDH1 in tumor ^a	1.174	0.882-1.562	0.272
ALDH1 in stroma ^a	1.548	1.078-2.222	0.018
Age group ^b	1.345	1.006-1.798	0.045
Differentiation grade ^c	1.381	1.043-1.828	0.024
TNM	1.699	1.047-2.758	0.032
Size ^d	1.482	0.969-2.266	0.690
Gender	1.371	0.850-2.211	0.196

RR: Relative risk; 95% CI: 95% confidence interval; ALDH1: aldehyde dehydrogenase-1. ^aALDH1-negative/weak group *versus* moderate expression group *versus* strong expression group; ^bage<50 years *versus* 50-60 years *versus* >60 years; ^cwell-differentiated *versus* moderately differentiated *versus* poorly differentiated; ^dsize <30 mm *versus* 30-60 mm *versus* >60 mm.

ALDH1-negative, moderate, and strong expression groups were 139, 118, and 65 months, respectively.

Out of the 138 cases available for survival analysis, ALDH1 expression in stromal cells was negative, moderate, and strong in 14, 46, and 78 cases, respectively. The median overall survival in the stromal ALDH1-negative, moderate and strong expression groups were 122, 98, and 76 months, respectively. ALDH1 expression in stromal cells was not significantly associated with overall survival by statistical analysis ($p=0.142$; Figure 2B).

Multivariate analysis was performed using the Cox Regression method based on the above clinicopathological parameters and ALDH1 expression in tumor and stromal cells (Table IV). Increasing age at diagnosis, TNM stage, differentiation grade and ALDH1 expression in stromal cells were independent risk factors for poorer overall survival in ESCC ($p<0.05$), while other variables such as tumor size, sex, and ALDH1 expression in tumor cells did not independently contribute to overall survival ($p>0.05$).

Discussion

In ESCC, as in other types of cancer, many factors play a role in treatment failure, including chemoresistance, radioresistance, relapse after surgery, and metastasis. CSCs were determined to be the cause of these issues, with increasing interest in CSCs in the past decade. CSCs represent only a small percentage of tumor cells but are equipped with the properties of tumor-initiation, self-renewal and differentiation (22). Additionally, they are resistant to both conventional chemotherapy and ionizing radiation, and are thus a major factor involved in tumor relapse, invasion, and metastasis (23, 24). Examining CSC populations in

individual patients could predict for metastatic progression and clinical outcome. Targeting CSCs should improve the survival of patients with cancer. Previous research has shown that the CSC phenotype may combine both its cell of origin and the oncogenic transforming events (25, 26). Therefore, focusing on conserved stem and progenitor cell functions is one approach in the search for shared CSC markers. ALDH1, responsible for the oxidation of intracellular aldehydes, has been put forward as a potentially reliable CSC marker.

ALDH1 is involved in regulating stem cell differentiation and proliferation, particularly through the retinoid signaling pathway (12, 27, 28). ALDH1 expression has been observed in healthy stem cells such as human hematopoietic and neural stem and progenitor cells (29, 30). Deng *et al.* reported that ALDH1 is variably expressed in normal tissues, for instance, it is absent or only moderately expressed in breast and lung tissue, somewhat weakly expressed in colonic and gastric epithelia, and strongly expressed in liver and pancreatic tissue (31). Yoshitaka *et al.* reported no expression of ALDH1 in normal oral squamous epithelial cells (32). However, no studies have reported on the expression of ALDH1 in esophagus epithelial cells to our knowledge.

ALDH1 has been shown to be related to the CSC phenotype. When Ginestier *et al.* implanted a very small number of ALDH1-positive cells into the fat pad of nude mice, a breast tumor formed (33). ALDH1 was later considered as a CSC marker in other types of human solid tumors, including ovarian (34, 35), lung (17, 36) and rectal cancer (37). The presence of cancer cells with such a stem cell marker was associated with a worse outcome. In some types of cancer, ALDH1 has been linked to chemoresistance and proposed as a potential therapeutic target (38, 39). However, the expression patterns of ALDH1 in malignant tumors are inconsistent. ALDH1 expression was associated with worse clinical outcomes in patients with breast cancer, while related to higher tumor grade in some other studies (40, 41). Liu *et al.* reported an association between ALDH1 expression and aggressive pathological features in gallbladder adenocarcinoma (18). Reports about ovarian cancer have been contradictory, with some studies showing ALDH1 expression to be related to poor overall survival (31) and others showing contrasting results (20). The mechanism responsible for such variations in ALDH1 expression in malignant tumors needs further investigation.

In the present study, we examined the ALDH1 expression patterns in a series of 157 ESCC samples. We found that ALDH1 protein expression in tumors was significantly related to poor histological differentiation, however, no significant association was found between the ALDH1 protein expression and other clinical parameters such as age, sex, tumor size, or TNM stage. Among the 138 ESCC cases available for overall survival analysis, strongly positive staining of ALDH1 protein in ESCC cancer cells was

associated with shorter overall survival times. These findings are consistent with previous studies. Ajani *et al.* investigated ALDH1 expression in 167 cases of esophageal adenocarcinoma and squamous cell carcinoma and reported high ALDH1 expression to be associated with therapy resistance and aggressive phenotype (42). Minato *et al.* examined 152 patients with ESCC and found that ALDH1 expression was a predictor of postoperative recurrence and prognosis (43). In a study of 79 cases of ESCC, Wang *et al.* found ALDH1 expression to be correlated with poor histological differentiation and poor 5-year OS compared to those for the negative group (44). In our study, there was no significant difference in ALDH1 expression by TNM staging. However, owing to the very limited number of stage I and stage IV samples in our study, we separated samples into two TNM groups: I/II and III/IV. This may explain why there is no clinical stage correlation with ALDH1 expression, which is in contrast with the findings reported by Wang *et al.* (44).

Previous studies have reported frequent and strong intra-stromal expression of ALDH1 in both non-malignant and tumor-associated stromal cells. However, the potential role of ALDH1 expression in the tumor microenvironment remains unclear. For example, some breast cancer researchers hold the opinion that high ALDH1 expression in tumor-related stromal cells predicts a better clinical outcome (13, 45), while others believe that ALDH1 expression in stromal cells is not associated with any clinical parameters (14, 46). In our study, we observed high expression levels of ALDH1 in ESCC stromal cells but determined that these levels had no significant association with the noted clinical parameters.

In summary, our results confirm that strong positive expression of ALDH1 is related to poor survival in ESCC. Moreover, the expression of ALDH1 is inversely correlated with histological differentiation. Our study also showed that there is no relationship between intra-stromal ALDH1 expression and standard clinical parameters. However, more studies are warranted to verify both the role of ALDH1 and the molecular mechanisms involved in the progression of ESCC.

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